

Date: September 23, 2026
To: Editor-in-Chief, *Computers & Fluids*
Elsevier Publishing

Subject: Submission of Original Research Article for Publication in *Computers & Fluids*

Dear Editor-in-Chief and Editorial Board,

We are pleased to submit our original research manuscript entitled:

“A High-Resolution Finite-Difference Benchmark for Steady Continuation and Hopf Bifurcation Dynamics up to Extreme Reynolds Numbers ($Re = 100,000$) in 2D Lid-Driven Cavity Flows”

for consideration as a regular research paper in *Computers & Fluids*.

Significance and Novelty of the Work:

The incompressible 2D lid-driven cavity flow remains the quintessential proving ground for computational fluid dynamics. However, established benchmarks have largely stagnated at $Re \leq 10,000$ (e.g., Ghia et al., 1982) or $Re \leq 25,000$ (Erturk et al., 2005) due to numerical divergence stemming from the lid corner singularities and escalating cell Péclet numbers ($Pe \gg 2$).

In this manuscript, we push the resolution and Reynolds frontiers of the standard finite-difference streamfunction-vorticity (ψ - ω) formulation through several key technical advances:

1. **Exact Spectral Poisson Acceleration:** We integrate an exact Discrete Sine Transform (DST-I) Poisson solver operating at optimal $\mathcal{O}(N^2 \log N)$ complexity, eliminating iterative convergence errors, pressure-velocity decoupling, and artificial numerical dissipation ($\nu_{\text{num}} = 0$).
2. **Asymptotic Batchelor Core Verification at $Re = 100,000$:** We compute steady continuation solutions up to an unprecedented $Re = 100,000$ on ultra-fine grids (513×513) and a mega-mesh (1025×1025 , 1,050,625 nodes). We quantitatively demonstrate convergence to Batchelor’s (1956) classical asymptotic core theorem: the primary vortex center migrates exactly to the geometric mid-plane ($x_c \rightarrow 0.50000$), while core vorticity stabilizes to an invariant plateau value of $\omega_0 \approx -2.54$.
3. **Interior Hybrid Differencing & Numerical Dissipation Mapping:** At $Re = 100,000$ ($Pe = 195.3 \gg 2$), an interior Spalding–Patankar hybrid differencing scheme regularizes corner singularity dispersion while maintaining well-bounded numerical dissipation in the core ($\bar{\nu}_{\text{num}}^{\text{excess}}/\nu \leq 0.99$ on 513×513 and 0.24 on 1025×1025), with pure second-order central differencing ($\nu_{\text{num}} = 0$) strictly preserved everywhere for all $Re \leq 50,000$.
4. **Unsteady Bifurcation Mapping on 513×513 Meshes:** Through physical time-accurate marching ($t = 16.0$ – 22.0), we systematically resolve the primary supercritical Hopf bifurcation ($Re = 10,000$, $St = 0.6249 \pm 0.0416$), modulated multi-frequency dynamics ($Re = 15,000$, $St = 0.5364 \pm 0.0278$), folded multi-frequency orbits ($Re = 25,000$, $St = 0.0953 \pm 0.0281$), boundary-layer vortex shedding ($Re = 50,000$, $St = 0.2581 \pm 0.0300$), and frontier high- Re shedding at $Re = 100,000$ ($St = 0.1908 \pm 0.0379$).
5. **Commitment to Open Science and Reproducibility:** We openly release 28 full 2D coordinate/flow field datasets, phase portrait scripts, and automated visualization tools under the MIT License with permanent Zenodo DOIs.

Suggested Potential Reviewers:

1. **Prof. Ercan Erturk** (Kocaeli University / ITU) — World authority on high-Reynolds lid-driven cavity benchmarking and steady Navier–Stokes solutions up to $Re = 25,000$.
2. **Prof. Olivier Botella** (Université de Lorraine, LEMTA) — Renowned specialist in high-order compact finite difference schemes and incompressible cavity benchmarks.
3. **Prof. Chang Shu** (National University of Singapore) — Pioneer of differential quadrature and lattice Boltzmann / FDM formulations for high-Reynolds cavity flows.
4. **Prof. Jie Shen** (Purdue University) — Expert in spectral methods, projection schemes, and Navier–Stokes stability.

Declarations:

This manuscript is an original work, has not been published previously, and is not under simultaneous consideration for publication elsewhere. All authors have read and approved the final manuscript. There are no competing financial or non-financial interests to declare.

Sincerely yours,

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